Yun-Chen "Jennifer" Hung

GitHub: https://github.com/yunchen-hung
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Portfolio: https://yunchen-hung.github.io/

Research Interests

Cognitive Modeling, NeuroAI, Sequential Memory Recall, Computational Neuroscience, Deep Learning

Education

Rutgers University–New Brunswick, New Brunswick, NJ — Sep 2025- Present

Ph.D., Cognitive Psychology — cGPA: 4.00 / 4.00
- **Advisor**: Dr. Qiong Zhang

University of California, San Diego (UCSD), La Jolla, CA — Sep 2021- June 2025

B.S., Data Science, *Magna Cum Laude* — cGPA: 3.95 / 4.00
- Data Science Capstone: Exploring Working Memory Task Predictability using fMRI Data.
  (**Advisor:** Dr. Armin Schwartzman)

B.S., Cognitive Science with a Specialization in Neuroscience, *Distinction in Cognitive Science*
- Cognitive Science Honors Thesis: Testing the Flexibility of Pupillary Working Memory Signals to Visual and Semantic Task Demands.
  (**Primary Advisor:** Dr. Anastasia Kiyonaga, **Secondary Advisor:** Dr. Tim Brady)

Research Experience

Graduate Research Assistant, Rutgers University

Memory Optimzation Lab, Dr. Qiong Zhang — Aug 2025 – Present
- Lead independent project on using deep neural networks to identify optimal human serial recall strategies.
- Trained recurrent neural networks with reinforcement learning and explored decoding of hidden states.

Undergraduate Research Assistant, UCSD

Kiyonaga Lab, Dr. Anastasia Kiyonaga — Feb 2024 – Aug 2025
- Investigated the relationship between visual Working Memory and pupil response/ oculomotor movements.
- Modified experiment on Psychopy, wrote scripts for probe generation and pupil data cleaning with Python.
- Proficiency in performing calibration of eye-tracking tools through collecting 100+ human subject data.

Undergraduate Research Assistant, UCSD

Serences Lab, Dr. John Serences — Mar 2023 – Jun 2024
- Devised experiment on unconscious serial dependence with Continuous Flash Suppression using Psychopy.
- Presented and hosted 3+ monthly cognitive neuroscience Journal Club.

Summer Intern, Neuroscience Program of Academia Sinica (NPAS)

Tung Lab, Dr. Yi-Chung Tung — Jul 2022 – Aug 2022
- Studied the migration and differentiation of neuroblastoma under an oxygen gradient.
- Presented research results in a presentation to 30+ summer interns and researchers of Academia Sinica.

Technical Experience

Machine Learning Intern, ActEarn

Wander Social App — Jun 2024 – Sep 2024
- Assisted in development of social app which matches energy-aligned individuals by their answers to given questions
- Developed an internal tool on React for executive team to easily edit and update 10,000+ questions and user answers stored in GCP Big Query and AWS DynamoDB, met engineering team weekly to meet demands
- Optimized database access speed for team members by 70%, worked with 5+ different APIs

Lab Assistant, UCSD

Qualcomm Institute Nano3 Cleanroom — Jan 2022 – Jan 2024
- Proficient in training 30+ Nano3 users on tools (Heidelberg MLA 150, ellipsometers and mask aligners).
- Experience in making samples for prominent companies with dry etching using the Oxford Plasmalab 80, microfabrication with the EVG 620 Mask Aligner, and thin metal deposition with Denton Discovery 635.

Teaching Experience

Teaching Assistant

Conduct exam review sessions, design homework, hold weekly office hours, grade assignments

Theoretical Foundations of Data Science II, *UCSD* — Spring 2024

Neuroanatomy and Physiology, *UCSD* — Fall 2023

Skills & Certifications

Programming Languages

Python, R, MATLAB, SQL, Java, HTML, CSS, LaTeX, C++, JavaScript, Svelte

Libraries

PyTorch, pandas, NumPy, Scikit-Learn, OpenAI Gym, Dask, React, SciPy, StatsModel

Programming Tools

Spark, AWS, Git/GitHub, VS Code, IntelliJ

Neuroscience Tools

Psychopy, Eye tracking (Eyelink), ImageJ, Centrifuge, Microscopy

Languages

Mandarin, English

Awards & Honors

- Excellence Fellowship, *Rutgers University* — 2025-26
- Robert Glushko '79 Cognitive Science Honors Thesis Award, *UCSD* — 2025
- Data Science Student Council Award, *UCSD* — 2025
- Data Science Community Award, *UCSD* — 2025
- Sponsorship to attend the annual Grace Hopper Celebration (GHC) (*$2,250*), *UCSD* — 2024
- Provost Honors, *UCSD* — 2021-2025

Publications

1. Dong, Y., **Hung, Y.-c.**, Xie, C., Kiyonaga, A., Windows to the goal: Pupillary working memory signatures prospectively adapt to task demands, iScience (2026), https:// doi.org/10.1016/j.isci.2026.115920.

2. Dong Y., **Hung Y**., Gaspariani L., Kiyonaga A. (2025). Gaze reinstatement during working memory for natural scenes, Journal of Neuroscience (2026), https://doi.org/10.1101/2025.09.29.678115

Poster Presentations

1. Dong, Y., **Hung, Y.**, Xie, C., Kiyonaga, A. (2025). Pupillary Signatures of Working Memory Content are Flexible to Visual vs. Semantic Task Demands. Poster presentation at Vision Sciences Society 2025.

2. **Hung, Y.**, Dong, Y., Brady, T., Kiyonaga, A. (2025). Testing the Flexibility of Pupillary Working Memory Signals to Visual and Semantic Task Demands. Presentation at Honors Thesis Presentation Conference 2025.

3. **Hung, J.**[1], Ancayan, J., Narayanan, S., Riegner, G., Schwartzman, A. (2025). Exploring Working Memory Task Predictability using fMRI Data. Poster presentation at 2025 HDSI Capstone Showcase.

Services & Outreach

Halıcıoğlu Data Science Institute (HDSI) Student Council — Aug 2024 – June 2025

Inaugural President

- Led 20+ weekly meetings with 15 student council members to plan events for students, and established first official HDSI graduation stole.
- Interviewed 2+ Dean candidates for the School of Computing, Information, and Data Science (SCIDS).
- Expressed concerns about the data science program in the undergraduate program review.

Data Science Student Society — Apr 2024 – Apr 2025

DataHacks Co-Director

- Facilitating a 200-person hackathon open to all majors and backgrounds.
- Connected with 20+ industry partners and the Halıcıoğlu Data Science Institute faculty for sponsorship.

[1] Went by preferred name of Jennifer Hung